习题 6-4

1. 求函数 $z=\dfrac{y}{x}$ 当 $x=2,\ y=1,\ \Delta x=0.1,\ \Delta y=-0.2$ 时的全增量和全微分.

解 由 $z_x=-\dfrac{y}{x^2},\quad z_y=\dfrac{1}{x}$，则对于函数 $z(x,y)=\dfrac{y}{x}$,

$$\Delta z=z(2+0.1,\ 1-0.2)-z(2,1)=\frac{1-0.2}{2+0.1}-\frac{1}{2}\approx -0.119,$$

$$\mathrm{d}z\big|_{\substack{x=2,y=1\\ \Delta x=0.1,\Delta y=-0.2}}=\left[-\frac{y}{x^2}\Delta x+\frac{1}{x}\Delta y\right]_{\substack{x=2,y=1\\ \Delta x=0.1,\Delta y=-0.2}}=-\frac{1}{4}\times 0.1+\frac{1}{2}\times(-0.2)=-0.125.$$

2.求下列函数的全微分.

(1) $z=xy+\dfrac{x}{y}$;　　(2) $z=\arcsin\dfrac{x}{y}$;

(3) $z=y^x$;　　(4) $u=e^x(x^2+y^2+x^2)$.

解（1）因 $z_x=y+\dfrac{1}{y},\quad z_y=x-\dfrac{x}{y^2}$，故

$$\mathrm{d}z=\left(y+\frac{1}{y}\right)\mathrm{d}x+\left(x-\frac{x}{y^2}\right)\mathrm{d}y.$$

（2）因 $z_x=\dfrac{\dfrac{1}{y}}{\sqrt{1-\left(\dfrac{x}{y}\right)^2}}=\dfrac{|y|}{y\sqrt{y^2-x^2}},\quad z_y=\dfrac{-\dfrac{x}{y^2}}{\sqrt{1-\left(\dfrac{x}{y}\right)^2}}=-\dfrac{x|y|}{y^2\sqrt{y^2-x^2}}$, 故

$$\mathrm{d}z=\frac{|y|}{y\sqrt{y^2-x^2}}\mathrm{d}x-\frac{x|y|}{y^2\sqrt{y^2-x^2}}\mathrm{d}y\,.$$

（3）因 $z_x=y^x\ln y,\quad z_y=xy^{x-1}$，故

$$\mathrm{d}z=y^x\ln y\mathrm{d}x+xy^{x-1}\mathrm{d}y.$$

（4）因 $u_x=\mathrm{e}^x\left(x^2+y^2+z^2\right)+2x\mathrm{e}^x,\quad u_y=2y\mathrm{e}^x,\quad u_z=2z\mathrm{e}^x$，故

$$\mathrm{d}u=\left[\mathrm{e}^x\left(x^2+y^2+z^2\right)+2x\mathrm{e}^x\right]\mathrm{d}x+2y\mathrm{e}^x\mathrm{d}y+2z\mathrm{e}^x\mathrm{d}z.$$

习题 6—5

1. 求下列复合函数的导数：

（1）设 $z=\arcsin\dfrac{x}{y}$，而 $y=\sqrt{x^2+1}$，求 $\dfrac{\mathrm{d}z}{\mathrm{d}x}$；

（2）设 $z=x^2y-xy^2$，而 $x=u\cos v$，$y=u\sin v$，求 $\dfrac{\partial z}{\partial u},\dfrac{\partial z}{\partial v}$；

（3）设 $z=f\left(x^2-y^2,\ \mathrm{e}^{xy}\right)$，求 $\dfrac{\partial z}{\partial x}$，$\dfrac{\partial z}{\partial y}$，$\dfrac{\partial^2 z}{\partial x^2}$；

（4）设 $F(x,\ x+y,\ x+y+z)=0$，求 $\dfrac{\partial z}{\partial x},\dfrac{\partial z}{\partial y}$.

解（1）$\dfrac{\mathrm{d}z}{\mathrm{d}x}=\dfrac{\partial z}{\partial x}+\dfrac{\partial z}{\partial y}\dfrac{\mathrm{d}y}{\mathrm{d}x}$

$$=\frac{1}{\sqrt{1-\left(\dfrac{x}{y}\right)^2}}\cdot\frac{1}{y}+\frac{1}{\sqrt{1-\left(\dfrac{x}{y}\right)^2}}\cdot\left(-\frac{x}{y^2}\right)\cdot\frac{1}{2\sqrt{x^2+1}}\cdot 2x$$

$$=\sqrt{x^2+1}\cdot\frac{1}{\sqrt{x^2+1}}+\sqrt{x^2+1}\cdot\left(-\frac{x^2}{x^2+1}\right)\cdot\frac{1}{\sqrt{x^2+1}}$$

$$=\frac{1}{x^2+1}.$$

（2） $\dfrac{\partial z}{\partial u}=\dfrac{\partial z}{\partial x}\cdot\dfrac{\partial x}{\partial u}+\dfrac{\partial z}{\partial y}\cdot\dfrac{\partial y}{\partial u}$

$$=(2xy-y^2)\cdot\cos v+(x^2-2xy)\cdot\sin v$$
$$=3u^2\cos^2 v\sin v-3u^2\sin^2 v\cos v.$$

$$\frac{\partial z}{\partial v}=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial v}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial v}$$
$$=(2xy-y^2)\cdot\left(-u\sin v\right)+(x^2-2xy)\cdot u\cos v$$
$$=-2u^3\sin v\cos v\left(\sin v+\cos v\right)+u^3\left(\sin^3 v+\cos^3 v\right).$$

（3）令 $u=x^2-y^2$，$v=\mathrm{e}^{xy}$，则 $z=f\left(u,\ v\right)$，故

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}=f_1'\cdot 2x+f_2'\cdot y\mathrm{e}^{xy}=2xf_1'+y\mathrm{e}^{xy}f_2',$$
$$\frac{\partial z}{\partial y}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial y}=f_1'\cdot\left(-2y\right)+f_2'\cdot x\mathrm{e}^{xy}=-2yf_1'+x\mathrm{e}^{xy}f_2',$$

而

$$\frac{\partial^2 z}{\partial x^2}=\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)=\frac{\partial}{\partial x}\left(2xf_1'+y\mathrm{e}^{xy}f_2'\right)=\frac{\partial}{\partial x}\left(2xf_1'\right)+\frac{\partial}{\partial x}\left(y\mathrm{e}^{xy}f_2'\right)$$

$$=\left(2f_1'+2x\frac{\partial f_1'}{\partial x}\right)+\left(y^2\mathrm{e}^{xy}f_2'+y\mathrm{e}^{xy}\frac{\partial f_2'}{\partial x}\right). \qquad (*)$$

其中，因 f_1', f_2' 仍保持与 f 相同的函数关系，故

$$\frac{\partial f_1'}{\partial x}=\frac{\partial f_1'}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial f_1'}{\partial v}\cdot\frac{\partial v}{\partial x}=f_{11}''\cdot 2x+f_{12}''\cdot y\mathrm{e}^{xy}=2xf_{11}''+y\mathrm{e}^{xy}f_{12}'',$$
$$\frac{\partial f_2'}{\partial x}=\frac{\partial f_2'}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial f_2'}{\partial v}\cdot\frac{\partial v}{\partial x}=f_{21}''\cdot 2x+f_{22}''\cdot y\mathrm{e}^{xy}=2xf_{21}''+y\mathrm{e}^{xy}f_{22}''.$$

将 $\frac{\partial f_1'}{\partial x}$，$\frac{\partial f_2'}{\partial x}$ 代入式（*），得

$$\begin{aligned}\frac{\partial^2 z}{\partial x^2}&=\left[2f_1'+2x\left(2xf_{11}''+y\mathrm{e}^{xy}f_{12}''\right)\right]+\left[y^2\mathrm{e}^{xy}f_2'+y\mathrm{e}^{xy}\left(2xf_{21}''+y\mathrm{e}^{xy}f_{22}''\right)\right]\\&=\left[2f_1'+4x^2f_{11}''+2xy\mathrm{e}^{xy}f_{12}''\right]+\left[y^2\mathrm{e}^{xy}f_2'+2xy\mathrm{e}^{xy}f_{21}''+y^2\mathrm{e}^{2xy}f_{22}''\right]\\&=2f_1'+y^2\mathrm{e}^{xy}f_2'+4x^2f_{11}''+4xy\mathrm{e}^{xy}f_{12}''+y^2\mathrm{e}^{2xy}f_{22}''.\end{aligned}$$

（4）令 $G(x,y,z)=F(x,\,x+y,\,x+y+z)=F(u,v,w)$，其中

$$u=x,\ v=x+y,\ w=x+y+z.$$

因

$$G_x=F_1'\cdot\frac{\partial u}{\partial x}+F_2'\cdot\frac{\partial v}{\partial x}+F_3'\cdot\frac{\partial w}{\partial x}=F_1'\cdot 1+F_2'\cdot 1+F_3'\cdot 1=F_1'+F_2'+F_3',$$
$$G_y=F_2'\cdot\frac{\partial v}{\partial y}+F_3'\cdot\frac{\partial w}{\partial y}=F_2'\cdot 1+F_3'\cdot 1=F_2'+F_3',$$
$$G_z=F_3'\cdot\frac{\partial w}{\partial y}=F_3'\cdot 1=F_3',$$

故

$$\frac{\partial z}{\partial x}=-\frac{G_x}{G_z}=-\frac{F_1'+F_2'+F_3'}{F_3'},\qquad \frac{\partial z}{\partial y}=-\frac{G_y}{G_z}=-\frac{F_2'+F_3'}{F_3'}.$$

2. 设 $z=x^2f\left(\frac{y}{x^2}\right)$，$f$ 为可微函数，证明 $x\frac{\partial z}{\partial x}+2y\frac{\partial z}{\partial y}=2z.$

证 令 $u=\frac{y}{x^2}$，则 $z=x^2f(u)$，故

$$\frac{\partial z}{\partial x}=\left(\frac{\partial z}{\partial x}\right)+\frac{\partial z}{\partial f}\cdot\frac{\mathrm{d}f}{\mathrm{d}u}\cdot\frac{\partial u}{\partial x}=2xf\left(u\right)+x^2\frac{\mathrm{d}f}{\mathrm{d}u}\cdot\left(-2\frac{y}{x^3}\right)=2xf\left(u\right)-\frac{2y}{x}\cdot\frac{\mathrm{d}f}{\mathrm{d}u},$$
$$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial f}\cdot\frac{\mathrm{d}f}{\mathrm{d}u}\cdot\frac{\partial u}{\partial y}=x^2\frac{\mathrm{d}f}{\mathrm{d}u}\cdot\frac{1}{x^2}=\frac{\mathrm{d}f}{\mathrm{d}u},$$

所以

$$x\frac{\partial z}{\partial x}+2y\frac{\partial z}{\partial y}=x\left[2xf\left(u\right)-\frac{2y}{x}\cdot\frac{\mathrm{d}f}{\mathrm{d}u}\right]+2y\frac{\mathrm{d}f}{\mathrm{d}u}$$
$$=2x^2f\left(u\right)-2y\frac{\mathrm{d}f}{\mathrm{d}u}+2y\frac{\mathrm{d}f}{\mathrm{d}u}$$
$$=2x^2f\left(u\right)=2z.$$

3.求下列隐函数的导数.

（1） $\sin y+\mathrm{e}^x-xy^2=0,$求$\frac{\mathrm{d}y}{\mathrm{d}x}$；

（2） $x+2y+2z-2\sqrt{xyz}=0$，求$\frac{\partial z}{\partial x},\frac{\partial z}{\partial y}$；

（3） $\mathrm{e}^z-xyz=0,$求$\frac{\partial^2 z}{\partial x^2},\frac{\partial^2 z}{\partial x\partial y}.$

解（1）令 $F(x,y)=\sin y+\mathrm{e}^x-xy^2$，则

$$F_x=\mathrm{e}^x-y^2,\ F_y=\cos y-2xy,$$

故$\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{F_x}{F_y}=\frac{\mathrm{e}^x-y^2}{2xy-\cos y}.$

（2）设 $F(x,y,z)=x+2y+2z-2\sqrt{xyz}$，则

$$F_x=1-2\cdot\frac{1}{2\sqrt{xyz}}\cdot yz=\frac{\sqrt{xyz}-yz}{\sqrt{xyz}},\quad F_y=2-2\frac{1}{2\sqrt{xyz}}\cdot xz=\frac{2\sqrt{xyz}-xz}{\sqrt{xyz}},$$
$$F_z=2-2\frac{1}{2\sqrt{xyz}}\cdot xy=\frac{2\sqrt{xyz}-xy}{\sqrt{xyz}},$$

于是

$$\frac{\partial z}{\partial x}=-\frac{F_x}{F_z}=-\frac{\sqrt{xyz}-yz}{2\sqrt{xyz}-xy},\qquad\frac{\partial z}{\partial y}=-\frac{F_y}{F_z}=-\frac{2\sqrt{xyz}-xz}{2\sqrt{xyz}-xy}.$$

（3） $F(x,y,z)=\mathrm{e}^z-xyz,$ 则

$$F_x=-yz,\quad F_y=-xz,\quad F_z=\mathrm{e}^z-xy,$$

于是

$$\frac{\partial z}{\partial x}=-\frac{F_x}{F_z}=\frac{yz}{\mathrm{e}^z-xy},\qquad \frac{\partial z}{\partial y}=-\frac{F_y}{F_z}=\frac{xz}{\mathrm{e}^z-xy}.$$

$$\begin{aligned}\frac{\partial^2 z}{\partial x^2}&=\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)=\frac{\partial}{\partial x}\left(\frac{yz}{\mathrm{e}^z-xy}\right)\\&=\frac{y\dfrac{\partial z}{\partial x}\cdot\left(\mathrm{e}^z-xy\right)-\left(\mathrm{e}^z\dfrac{\partial z}{\partial x}-y\right)\cdot yz}{\left(\mathrm{e}^z-xy\right)^2}=\frac{2y^2z\mathrm{e}^z-2xy^3z-y^2z^2\mathrm{e}^z}{\left(\mathrm{e}^z-xy\right)^3}.\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 z}{\partial x\partial y}&=\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)=\frac{\partial}{\partial y}\left(\frac{yz}{\mathrm{e}^z-xy}\right)\\&=\frac{\left(z+y\dfrac{\partial z}{\partial y}\right)\cdot\left(\mathrm{e}^z-xy\right)-\left(\mathrm{e}^z\dfrac{\partial z}{\partial y}-x\right)\cdot yz}{\left(\mathrm{e}^z-xy\right)^2}=\frac{z\mathrm{e}^{2z}-xyz^2\mathrm{e}^z-x^2y^2z}{\left(\mathrm{e}^z-xy\right)^3}.\end{aligned}$$